



# Proposed CEA Link — December 2026

Weeks 17-18 Swapped re-ordered by hand to cut fatigue — same duties, same days, same coverage; only which week sits next to which has changed

Prepared 22 September 2026 · re-rendered 24 September 2026 · built for the December 2026 timetable and assessed by the workspace's own rule modules (Marylbone Roster v24.22) · figures on this page are computed, not typed

## PROPOSAL

# Targeted Fatigue Redo

Weeks 17-18 Swapped re-ordered by hand — same duties, same days, same coverage

**CODE** TF-24-EXT

**FINGERPRINT** 8eef9a13

**BUILT FROM** duty table — · seed — · 24 lines · 4 cover weeks

**LINEAGE** Supplied as a Word table and assessed here — not produced by the proposal search. It is a pure RE-ORDERING of "Weeks 17-18 Swapped" (WS-24-EXT · 0bebb675): identical duty table, identical per-day duty multiset and an identical hour-by-hour coverage curve, with only the order of the weeks changed. Compare with the searched proposals "Weekday Lates 2/3/4", which are the same kind of work done by the annealer.

✓ 24 lines designed

✓ 35h 00m a week — exactly the contract

✓ 0 hard-limit breaches

✓ 0 rests under 12h

⚠ 4 fatigue factors present — 3 as rostered (today: 5)

## 13 of 19

shift times are today's

6 times are new; the rest are turns people already work

## 9 days

longest run of worked days

today's link reaches 9; Chiltern's limit is 13

## 4 of 20

working weeks are one turn

same clock time all week, as 7 of today's 16 are

## 5 in 24

full weekends off

21% of the rotation, against 20% today (4 in 20)

## 4 cover weeks

lines 1, 7, 12, 18

gaps of 7, 6, 5, 6 around the wheel — close to even, one line out; unchanged from the design this re-orders

## 4h 10m a week

typical week-to-week move

how far the working day shifts each week — today's link moves 4h 0m a week

## What this is

A proposal for the CEA link once the December 2026 timetable arrives, for a team of 24 rather than 20. The brief was to keep the **feel** of the link people work today — a week is one turn, rest days come in pairs, four whole cover weeks spread round the rotation, and the same 06:20 / 08:00 / 14:00 / 15:15 clock times — while meeting the staffing shape agreed for December 2026 (four to open every day, three through to the close, four on a Saturday, five still on at 22:00, fourteen on a Saturday, ten on a Sunday) and every rule the link is assessed against: Chiltern's 13-day limit, twelve hours between duties, the exact 35-hour contracted week, and the ORR's good-practice fatigue factors.

It was not drawn by this app. It arrived as a Word table — a proposal from outside the workspace — and was read straight into the app's own shape, then judged by the same modules that judge a design the workspace generated itself: runDesignChecks, weeklyHours, assessHardLimits, assessFatigue, scoreOrder and calcHourlyCoverage. There is no search behind it and nothing to reproduce; what can be checked is every figure on these pages, each computed from the cells on page 7. Paste those cells into Links → Import and the workspace restates all of it.

**What it is not.** Not a recommendation and not a finished roster: nothing here has been through the roster office or a rep. The 24-line length, the four cover weeks and the December 2026 headcounts are owner-relayed figures with no document behind them (evidence class C), and the 13-day limit's policy citation is still outstanding — pages 6 and 7 say so where it applies. It clears every rule the tool can check, which is the floor for being worth discussing.

# The decision this sheet supports

What is being decided, which criteria are hard and which are the room's, how much each figure can bear, and what no page here can tell you. Read this before the tiles.

## The question

Should the link for the December 2026 timetable be the hand-built Weekday Lates line, in this revision, accepting 12 on a Saturday against the agreed 14 and 0 of the four December 2026 headcount rules met, for a week shape its authors chose?

**Targeted Fatigue Redo** is read two ways: against **Quarter To (QT-24-Q34)** and **By the Book (BB-24-D7)**, the two searched answers; and within its own line, against *Weeks 17-18 Swapped*, the parent whose changed cells are outlined on the next page.

## The criteria, in three tiers

### HARD — A DESIGN MEETS THESE OR CANNOT BE RUN

- ✓ Twelve hours between duties — met, tightest 13h 35m
- ✓ Chiltern's 13-day limit — longest run 9 days
- ✓ The 35-hour contract, exactly — 35h 00m

### SOFT — THE ROOM WEIGHS THESE AGAINST EACH OTHER

- December 2026 headcounts met: **0 of 4** (today 0 of 4)
- Fatigue factors present: **4** worst case, 3 as rostered (today 5)
- Demand fit weekday · Sat · Sun: **34.4** · 20.7 · 62.4 (today 44.7 · 45.7 · 65.9; lower is closer)
- Times worked today: **13 of 19** · full weekends off: **5** in 24 (today 4 in 20)

### PREFERENCE — STAFF'S TO STATE, NOT THE TOOL'S

- Early against late: earlies 7h 15m–8h 30m, lates 7h 30m–9h 10m
- The 6 new times: 07:00–15:30, 08:00–16:00, 08:30–17:00, 10:30–19:00, 12:00–20:30, 16:25–23:55
- The longest duty, 9h 10m, and who holds it
- Cover weeks at lines 1, 7, 12, 18

## How much each figure can bear

| FIGURE ON THESE PAGES                                      | WHERE IT COMES FROM                                                                          | WEIGHT                                      |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------|
| 0 rests under 12h · 35h 00m contract · 12/9 at the weekend | Computed from the cells; re-computed by the workspace on import                              | <b>Firm</b>                                 |
| Longest run 9 days, inside 13                              | Computed; the 13-day limit's policy citation is still outstanding                            | <b>Firm figure</b> , unconfirmed threshold  |
| Four at the open, three to close, 14 and 10 at the weekend | Owner-relayed December 2026 headcounts with no document behind them (class C)                | <b>Provisional</b>                          |
| 4 fatigue factors present                                  | The ORR's 2021 list, which it states are not limits; 4 factors still "definition to confirm" | <b>Advisory only</b> — never pass or fail   |
| Demand fit 34.4 · 20.7 · 62.4                              | Measured timetable, one share-based measure on one December 2026 timetable curve             | <b>Indicative</b>                           |
| 13 of 19 times are today's                                 | Computed against the live 20-line link                                                       | <b>Firm</b> , but a proxy for acceptability |
| 6 cells changed against Weeks 17-18 Swapped                | Computed cell by cell; outlined on the next page                                             | <b>Firm</b>                                 |

## What no page here can tell you

- Whether the people who would work them accept the times nobody works today (07:00–15:30, 08:00–16:00, 08:30–17:00, 10:30–19:00, 12:00–20:30, 16:25–23:55) — familiarity is measured as "times worked today", which is a proxy for acceptability, not acceptability.
- Whether the roster clerk splits a cover week day-on-day-off. It is the one thing that moves the 48-hour-break figure (12 as a block, 15 split), and the link does not decide it.
- How leave and sickness cover land on a 24-line link with four cover weeks (lines 1, 7, 12, 18); nothing here models absence.
- Whether the December 2026 timetable demand curve holds on Wembley event days — the Saturday weighting approximates them, and the curve is a measured timetable, not a footfall count.

## Before it is frozen

Lines 2, 6, 9, 16, 17 hold the times nobody works today; the people who would work those weeks read them first and say in one line each what they would change. The roster office answers the cover-week question. Someone confirms the source of the 13-day limit. Only then does the factor count on page 7 mean anything, and only then is this sheet more than an aid to a conversation.

## THE ROTATION

## The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

| LINE    | SUN            | MON            | TUE            | WED            | THU            | FRI            | SAT            | MON-SAT<br>hours | ALL WEEK<br>hours | DAYS WORKED<br>Mon-Sat | TURNS<br>Mon-Fri |
|---------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|-------------------|------------------------|------------------|
| 1       | SP             | SP             | SP             | SP             | SP             | SP             | SP             | 35h 00m          | 35h 00m           | —                      | —                |
| 2       | 14:30<br>23:25 | 16:25<br>23:55 | 16:25<br>23:55 | 16:25<br>23:55 | 16:25<br>23:55 | RD             | RD             | 30h 00m          | 38h 55m           | 4                      | <b>1</b>         |
| 3       | 07:15<br>15:45 | RD             | 08:30<br>16:30 | 06:20<br>14:20 | 06:20<br>14:20 | 06:20<br>14:20 | 06:20<br>14:20 | 40h 00m          | 48h 30m           | 5                      | 2                |
| 4       | RD             | RD             | 06:20<br>14:20 | 08:30<br>16:30 | RD             | 13:30<br>22:00 | 14:45<br>23:55 | 33h 40m          | 33h 40m           | 4                      | 3                |
| 5       | 14:30<br>23:25 | RD             | 14:00<br>22:30 | 14:00<br>22:30 | 14:00<br>22:30 | 16:25<br>23:55 | RD             | 33h 00m          | 41h 55m           | 4                      | 2                |
| 6       | RD             | 06:20<br>14:20 | 07:00<br>15:30 | 07:00<br>15:30 | 07:00<br>15:30 | 08:30<br>17:00 | RD             | 42h 00m          | 42h 00m           | 5                      | 3                |
| 7       | SP             | SP             | SP             | SP             | SP             | SP             | SP             | 35h 00m          | 35h 00m           | —                      | —                |
| 8       | 07:15<br>15:45 | 06:20<br>13:45 | RD             | RD             | 13:30<br>22:00 | 14:00<br>22:30 | 13:30<br>21:00 | 31h 55m          | 40h 25m           | 4                      | 3                |
| 9       | RD             | 12:00<br>20:30 | 12:00<br>20:30 | 12:00<br>20:30 | 12:00<br>20:30 | 12:00<br>20:30 | RD             | 42h 30m          | 42h 30m           | 5                      | <b>1</b>         |
| 10      | RD             | 14:00<br>22:30 | 16:25<br>23:55 | 16:25<br>23:55 | 16:25<br>23:55 | 16:25<br>23:55 | 14:45<br>23:55 | 47h 40m          | 47h 40m           | 6                      | 2                |
| 11      | RD             | 07:00<br>15:30 | 07:00<br>15:30 | RD             | RD             | 06:20<br>13:35 | 06:20<br>14:50 | 32h 45m          | 32h 45m           | 4                      | 2                |
| 12      | SP             | SP             | SP             | SP             | SP             | SP             | SP             | 35h 00m          | 35h 00m           | —                      | —                |
| 13      | RD             | RD             | 14:00<br>22:30 | 13:30<br>22:00 | 14:00<br>22:30 | 14:00<br>22:30 | 14:30<br>22:00 | 41h 30m          | 41h 30m           | 5                      | 2                |
| 14      | RD             | 16:25<br>23:55 | 16:25<br>23:55 | RD             | RD             | 07:00<br>15:30 | 06:20<br>14:50 | 32h 00m          | 32h 00m           | 4                      | 2                |
| 15      | 07:15<br>15:45 | 06:20<br>13:45 | 06:20<br>13:35 | 06:20<br>13:35 | 06:20<br>13:35 | 07:00<br>15:30 | RD             | 37h 40m          | 46h 10m           | 5                      | 3                |
| 16      | RD             | 08:30<br>17:00 | 08:30<br>17:00 | 08:30<br>17:00 | RD             | RD             | RD             | 25h 30m          | 25h 30m           | 3                      | <b>1</b>         |
| 17      | 10:30<br>19:00 | 16:25<br>23:55 | RD             | RD             | 06:20<br>14:20 | 06:20<br>14:20 | 06:20<br>14:50 | 32h 00m          | 40h 30m           | 4                      | 2                |
| 18      | SP             | SP             | SP             | SP             | SP             | SP             | SP             | 35h 00m          | 35h 00m           | —                      | —                |
| 19      | 14:30<br>23:25 | RD             | 06:20<br>13:45 | 06:20<br>13:45 | 06:20<br>13:45 | 06:20<br>13:45 | 08:00<br>16:00 | 37h 40m          | 46h 35m           | 5                      | 1                |
| 20      | 08:30<br>16:30 | 06:20<br>13:45 | RD             | RD             | 13:30<br>22:00 | 16:25<br>23:55 | 13:30<br>22:00 | 31h 55m          | 39h 55m           | 4                      | 3                |
| 21      | 13:30<br>22:00 | 13:30<br>22:00 | RD             | 13:30<br>22:00 | 13:30<br>22:00 | 13:00<br>21:00 | RD             | 33h 30m          | 42h 00m           | 4                      | 2                |
| 22      | RD             | 14:00<br>22:30 | 13:30<br>22:00 | 13:30<br>22:00 | 13:30<br>22:00 | RD             | RD             | 34h 00m          | 34h 00m           | 4                      | 2                |
| 23      | RD             | RD             | RD             | 16:25<br>23:55 | 16:25<br>23:55 | 16:25<br>23:55 | 16:25<br>23:55 | 30h 00m          | 30h 00m           | 4                      | <b>1</b>         |
| 24      | RD             | RD             | 06:20<br>13:45 | 06:20<br>13:45 | 07:00<br>15:30 | RD             | 06:20<br>13:45 | 30h 45m          | 30h 45m           | 4                      | 2                |
| Cover   | 9<br>SP 4      | 13<br>SP 4     | 15<br>SP 4     | 15<br>SP 4     | 16<br>SP 4     | 16<br>SP 4     | 12<br>SP 4     | 35h 00m          | 38h 13m           | 4.35                   | 4<br>one turn    |
| Minutes | 4,635          | 6,225          | 7,205          | 7,205          | 7,780          | 7,690          | 5,895          | 42,000           | 46,635            |                        |                  |

Early turns Late turns Cover (spare) week — four duties of seven, any turn Rest day

☒ Changed against *Weeks 17-18 Swapped (WS-24-EXT)* — 6 cells on 6 lines

Mon-Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines · Turns: distinct clock times Monday to Friday, **bold** where the week is one turn (one clock time Mon-Fri and one early-or-late family across every worked day — the page 1 figure) · Minutes: duty minutes per day, and Mon-Sat is the contract (20 × 35h = 42,000)

# Same feel, more people

What this design keeps of today, and where the December 2026 headcounts differ

## Kept — the shape of a week

| MEASURE                                                  | TODAY (20)          | PROPOSED (24)             |
|----------------------------------------------------------|---------------------|---------------------------|
| Working weeks that are one turn (one clock time Mon–Fri) | 7 of 16             | <b>4 of 20</b>            |
| Weeks mixing early and late turns                        | 7                   | <b>6</b>                  |
| Rest-day breaks that are two days or more                | 13 of 17            | <b>15 of 25</b>           |
| Single rest days between duties (not a 48h break)        | 4                   | <b>10</b>                 |
| Days worked in a week (lines × days)                     | 1×3, 3×4, 11×5, 1×6 | <b>1×3, 6×4, 9×5, 4×6</b> |
| Cover (spare) weeks (whole weeks)                        | lines 1, 7, 12, 17  | <b>lines 1, 7, 12, 18</b> |
| Distinct shift times (all but 6 on today's roster)       | 18                  | <b>19</b>                 |
| Mon–Sat hours, average week (the contract)               | 35h 00m             | <b>35h 00m</b>            |
| Sunday duties (not contracted; half the working lines)   | 8 of 16 lines       | <b>9 of 20 lines</b>      |

## Changed — the December 2026 headcount

| PEOPLE ON DUTY                                            | TODAY     | PROPOSED           | WHY                                                                     |
|-----------------------------------------------------------|-----------|--------------------|-------------------------------------------------------------------------|
| Monday to Friday                                          | 11–12     | <b>13–16</b>       | the contract: 20 working lines × 35h has to be worked somewhere         |
| Saturday                                                  | 10        | <b>12</b>          | owner's figure for December 2026, leaning late for events               |
| Sunday                                                    | 8         | <b>9</b>           | owner's figure for December 2026                                        |
| Opening at 06:20 (07:15 Sunday) wk · Sat · Sun            | 4 · 4 · 3 | <b>4 · 5 · 3</b>   | four to open, every day — not met: Saturday 5, Sunday 3                 |
| Through to the close (23:55; 23:25 Sunday) wk · Sat · Sun | 2 · 2 · 3 | <b>3–4 · 3 · 3</b> | three to close, four on a Saturday — not met: Fri 4, Saturday 3         |
| Still on duty at 22:00 wk · Sat · Sun                     | 4 · 2 · 3 | <b>4–6 · 3 · 3</b> | five from 22:00, Monday to Saturday — not met: Wed 4, Fri 6, Saturday 3 |

## The duty table, in today's times

| TURN                                  | WK           | SAT | SUN          | WK           | SAT          | SUN  |
|---------------------------------------|--------------|-----|--------------|--------------|--------------|------|
| 06:20–13:35 7h 15m · Early            | 1            |     | 1            |              |              |      |
| 06:20–13:45 7h 25m · Early            | 1            |     | 3 +2         | 1 +1         |              |      |
| <del>06:20–14:00 7h 40m · Early</del> |              | 4   |              | <del>4</del> |              |      |
| 06:20–14:20 8h 00m · Early            | 2            |     | 2            | 1 +1         |              |      |
| 06:20–14:50 8h 30m · Early            |              | 3   |              | 3            |              |      |
| 07:00–15:30 8h 30m · Early            |              |     | 2 +2         |              |              |      |
| 07:15–15:45 8h 30m · Early            |              |     | 3            |              |              | 3    |
| 08:00–16:00 8h 00m · Early            |              |     |              | 1 +1         |              |      |
| <del>08:00–16:30 8h 30m · Early</del> | <del>2</del> |     | <del>2</del> |              |              |      |
| 08:30–16:30 8h 00m · Early            |              |     | 1            | 1 +1         |              | 1    |
| 08:30–17:00 8h 30m · Early            |              |     |              | 1 +1         |              |      |
| 10:30–19:00 8h 30m · Early            |              |     |              |              |              | 1 +1 |
| <del>11:00–19:30 8h 30m · Late</del>  | <del>4</del> |     | <del>4</del> |              |              |      |
| 12:00–20:00 8h 00m · Late             |              | 4   |              | <del>4</del> |              |      |
| 12:00–20:30 8h 30m · Late             |              |     | 1 +1         |              | <del>4</del> |      |
| 13:00–21:00 8h 00m · Late             |              |     | 1            | 1 +1         |              | -1   |
| 13:30–21:00 7h 30m · Late             |              | 1   |              | 1            |              |      |
| 13:30–22:00 8h 30m · Late             | 1            |     | 4 +3         | 1 +1         | 1 +1         |      |
| 14:00–22:30 8h 30m · Late             | 2            |     | 2            |              |              |      |
| 14:30–22:00 7h 30m · Late             |              | 2   |              | 1 -1         |              |      |
| 14:30–23:25 8h 55m · Late             |              |     | 3            |              |              | 3    |
| 14:45–23:55 9h 10m · Late             |              | 2   |              | 2            |              |      |
| <del>15:15–23:55 8h 40m · Late</del>  | <del>2</del> |     | <del>2</del> |              |              |      |
| 16:25–23:55 7h 30m · Late             |              |     | 4 +4         | 1 +1         |              |      |

Busiest weekday · Saturday · Sunday. The first three columns are what the 20-line link does now; the last three are the proposal. Green cells grew, amber shrank, a struck-through row is a time the proposal does not use.

Why the duties land where they do: Monday to Saturday totals exactly 42,000 minutes, which is 20 × 35h to the minute, so the contract is paid. But unlike a table built day-uniform, **the weekdays here are not equal** — Mon 6,225 · Tue 7,205 · Wed 7,205 · Thu 7,780 · Fri 7,690, and Saturday 5,895. Monday is the lightest day by some margin and Thursday the heaviest; that is a property of the design rather than a fault, but it is the first thing to put to the roster office, because it decides how much cover each day really has.

# People on duty, hour by hour

Cover today and proposed against the measured December 2026 timetable (arrivals and departures, weighted by train length). A darker orange hour carries more of the day's traffic.

## Monday to Friday window 06:20–23:55 · one row per distinct weekday

|                       | 05 | 06 | 07  | 08  | 09  | 10 | 11 | 12 | 13 | 14 | 15 | 16  | 17  | 18  | 19  | 20 | 21 | 22 | 23 | FIT  |
|-----------------------|----|----|-----|-----|-----|----|----|----|----|----|----|-----|-----|-----|-----|----|----|----|----|------|
| TODAY MON · WED       |    | 4  | 4   | 6   | 6   | 6  | 7  | 7  | 7  | 7  | 7  | 7   | 5   | 5   | 5   | 4  | 4  | 4  | 2  | 55.8 |
| TODAY TUE · THU · FRI |    | 4  | 4   | 6   | 6   | 6  | 7  | 7  | 8  | 8  | 8  | 8   | 6   | 6   | 6   | 5  | 5  | 4  | 2  | 50   |
| PROPOSED MON          |    | 4  | 5   | 6   | 6   | 6  | 6  | 7  | 8  | 7  | 6  | 8   | 7   | 7   | 7   | 7  | 6  | 5  | 3  | 28.4 |
| PROPOSED TUE          |    | 4  | 6   | 8   | 8   | 8  | 8  | 9  | 10 | 9  | 8  | 9   | 7   | 7   | 7   | 7  | 6  | 5  | 3  | 43.6 |
| PROPOSED WED          |    | 4  | 5   | 7   | 7   | 7  | 7  | 8  | 11 | 9  | 8  | 10  | 8   | 8   | 8   | 8  | 7  | 4  | 3  | 42.3 |
| PROPOSED THU          |    | 4  | 6   | 6   | 6   | 6  | 6  | 7  | 11 | 11 | 9  | 10  | 10  | 10  | 10  | 10 | 9  | 5  | 3  | 48.8 |
| PROPOSED FRI          |    | 4  | 6   | 7   | 7   | 7  | 7  | 8  | 10 | 10 | 8  | 10  | 9   | 9   | 9   | 9  | 7  | 6  | 4  | 31   |
| DEC 2026 TRAFFIC      | 5  | 69 | 107 | 127 | 122 | 95 | 85 | 75 | 84 | 79 | 87 | 105 | 140 | 138 | 107 | 90 | 73 | 75 | 66 |      |

## Saturday window 06:20–23:55

|                  | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | FIT  |
|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------|
| TODAY            |    | 4  | 4  | 4  | 4  | 4  | 4  | 5  | 6  | 9  | 6  | 6  | 6  | 6  | 6  | 5  | 4  | 2  | 2  | 45.7 |
| PROPOSED         |    | 5  | 5  | 6  | 6  | 6  | 6  | 6  | 8  | 10 | 6  | 6  | 6  | 6  | 6  | 6  | 5  | 3  | 3  | 20.7 |
| DEC 2026 TRAFFIC | 6  | 24 | 58 | 75 | 81 | 92 | 88 | 77 | 64 | 68 | 72 | 84 | 82 | 67 | 64 | 79 | 78 | 62 | 41 |      |

## Sunday window 07:15–23:25

|                  | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | FIT  |
|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------|
| TODAY            |    |    | 3  | 4  | 4  | 4  | 4  | 4  | 5  | 8  | 8  | 5  | 4  | 4  | 4  | 4  | 3  | 3  | 3  | 65.9 |
| PROPOSED         |    |    | 3  | 4  | 4  | 5  | 5  | 5  | 6  | 9  | 9  | 6  | 5  | 5  | 4  | 4  | 4  | 3  | 3  | 62.4 |
| DEC 2026 TRAFFIC |    |    | 15 | 45 | 65 | 71 | 74 | 71 | 61 | 55 | 64 | 69 | 78 | 73 | 75 | 52 | 71 | 65 | 50 |      |

Spare cover is not in these figures — a cover week carries no times, so the rows are a floor. **Fit** is the squared gap between each hour's share of the day's traffic and its share of the cover, inside the day's window — lower is better, and today's link scores what it scores. It is measured on duty minutes, so an hour a duty only touches counts in proportion; the cells above count heads. It is the measure the searched tables were chosen on, and the fit of the average weekday is the *Wk fit* on page 8. Sunday: 5 December 2026 timetable movements fall after the 23:25 finish — the standing question on whether Sunday's window moves; the window is stored per design, so this proposal can be rebuilt to either answer.

**Reading it.** The proposal raises the whole day rather than reshaping it, which is what the December 2026 timetable brief asked for — the station rosters posts (ticket office, gateline, passenger assist) plus break cover, and a post needs someone on it whether the hour carries 23 trains or 5. The weekday's two peaks are a duty length apart, so cover stays fairly flat between them by arithmetic, not neglect.

# The checks sheet

Hard limits first — met or not. Then the ORR good-practice fatigue factors, which report what is present and never pass or fail a design.

**Hard limits** — a design either meets these or cannot be run

- ✓ **More than 13 consecutive days worked** — longest possible run **9** days (today: 9)  
Longest possible run is 9 days, within the 13 configured here from Chiltern practice (origin: the legacy Hidden standard). A spare week is 4 duties of 7 and the roster clerk places them, so this is the worst case the link allows. Worked as a BLOCK of four rather than split day-on-day-off, the answer is the same — **9** days — so this row does not depend on how a cover week is placed.  
Basis: Chiltern roster policy, citation outstanding — legacy Hidden 13-in-14 standard. Configured from Chiltern practice; the policy citation is outstanding, so this is stated as the app states it.
- ✓ **At least 12 hours between duties** — **0** rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today: 0)  
Tightest anywhere: **13h 35m — line 20 Fri 16:25-23:55 into Sat 13:30-22:00** (today's tightest: 12h 30m). The generator refuses a design it cannot repair to this; this design was checked against it, not generated.
- ✓ **The contracted week, exactly** — **35h 00m** average Mon-Sat over the 24 lines, cover weeks counted as contracted weeks  
700h of duty a week across 20 working lines. Sundays (77.25h) sit on top as RDW, as they do today. Individual weeks range 25h 30m–47h 40m; only the average is the contract.

**December 2026 timetable design figures** — the staffing shape agreed for the new timetable

| RULE                                           | PROPOSAL                                          |                                                                                                       |
|------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Four on at the open, every day                 | × 4 weekday · 5 Saturday · 3 Sunday               |                                                                                                       |
| Three through to the close; four on a Saturday | × 3–4 weekday · 3 Saturday · 3 Sunday             |                                                                                                       |
| Five still on duty at 22:00                    | × 4–6 weekday · 3 Saturday                        | a 22:00 finish is not "on at 22:00"                                                                   |
| Fourteen on a Saturday, ten on a Sunday        | × 12 · 9                                          |                                                                                                       |
| Cover weeks, evenly spread                     | × 4 weeks at lines 1, 7, 12, 18 — gaps 7, 6, 5, 6 | the December 2026 staffing shape asks for four, evenly spread — the count is right; the spread is not |
| About 4.2 days a week worked, Mon-Sat          | ✓ 4.35 over the 20 working lines                  |                                                                                                       |
| No :05 or :10 times except the open and close  | ✓ none                                            |                                                                                                       |
| Late turns slightly shorter than most earlies  | × not met — lates 7h30–9h10, earlies 7h15–8h30    | the longest late is longer than the longest early                                                     |

## To settle before it is frozen

**One of the two targets was hit outright; the other depends on how a cover week is worked.** The document sets out to clear two flags. The first is cleared on any reading: the worst seven-day total falls from **60.6 hours to 54.7**, under the 55-hour row, and the document's own estimate of "about 54h40" is right. The second is the interesting one. It reports the run "dropping from about 15 duties to 9", and there are three different measures in that sentence. **9** is the longest run of consecutive worked DAYS — correct, and inside Chiltern's thirteen. **FF11** counts shifts between **48-hour** breaks, and a single rest day is not one. On that measure this design is at **15 in the worst case**, down from 16 and over the threshold of 13 — but **12 as rostered**, under it. The difference is entirely the cover week: the 15 needs one split day-on-day-off so that it supplies no break and joins the blocks either side, and a cover week worked as a block always supplies one. **So whether this design clears FF11 is a question for the roster office, not for the link**, and page 7 prints both. **What the re-ordering cost.** Factors present fall from five to four in the worst case (three as rostered), but not uniformly: FF19 rises from **7 to 8**, full weekends off fall from **6 to 5**, and single-turn weeks fall from **6 of 20 to 4**, so more weeks now ask somebody to hold more than one start time. **Every hard gate stays clear** — 35h to the minute, no rest under 12 hours, no hard-limit breach, longest run nine days inside the thirteen on either reading. **The unfinished question is the same one the design it re-orders left open.** None of what remains needs a duty to change: FF11, FF15, FF8b and FF19 are all properties of which week sits next to which. On this exact duty table, the same shape work done by the workspace's own search takes the factors to **one** and the run to **six or seven** with the coverage curve untouched — the three searched proposals opposite are that result. This one is listed as supplied, deliberately, so the hand-made baseline stays on the record.

# ORR fatigue factors

Good practice guidelines — Fatigue Factors, p3 (December 2021).  $\Delta$  present · ✓ clear · ● standing, a property of the operation · – not applicable. Nathan assesses against this list.

| CODE | FACTOR                                                                                | TODAY'S LINK                                | PROPOSED                                     |
|------|---------------------------------------------------------------------------------------|---------------------------------------------|----------------------------------------------|
| FF1  | Night shift covering 00:00–05:00 Time of day                                          | --                                          | --                                           |
| FF2  | Early shift starting 05:00–07:00 Time of day                                          | ● 24                                        | ● 25                                         |
| FF3  | Very early shift starting before 05:00 Time of day                                    | ✓ 0                                         | ✓ 0                                          |
| FF4  | Very early shift (before 05:00) over 8h Duty length                                   | --                                          | --                                           |
| FF5  | Day shift over 12h Duty length                                                        | ✓ 0                                         | ✓ 0                                          |
| FF7  | Early shift over 10h Duty length                                                      | ✓ 0                                         | ✓ 0                                          |
| FF8b | Fewer than 2 days rest after a block of early starts Recovery time                    | $\Delta$ 3                                  | $\Delta$ 5                                   |
| FF11 | More than 13 consecutive shifts without a 48h break Recovery time                     | ✓ 12<br>✓ 9 as rostered                     | $\Delta$ 15<br>✓ 12 as rostered              |
| FF10 | More than 4 consecutive 12h day shifts Cumulative                                     | ✓ 0                                         | ✓ 0                                          |
| FF15 | More than 4 consecutive early shifts in a rotating pattern Cumulative                 | $\Delta$ 6                                  | $\Delta$ 7                                   |
| MRSF | More than 12 consecutive day shifts Cumulative                                        | ✓ 9                                         | ✓ 9                                          |
| MRSF | More than 55 hours worked in any 7-day period Cumulative                              | $\Delta$ 60.8                               | ✓ 54.7                                       |
| MRSF | More than 7 consecutive 8h shifts (definition to confirm) Cumulative                  | $\Delta$ 9                                  | ✓ 7                                          |
| MRSF | More than 6 consecutive night or early shifts in a permanent pattern Cumulative       | --                                          | --                                           |
| FF17 | Backward rotating pattern (definition to confirm) Circadian                           | ✓ 10 backward / 12 forward                  | ✓ 12 backward / 17 forward                   |
| FF18 | Rotating pattern of about a week (definition to confirm) Circadian                    | ● 20-line rotation · typically 4h 0m a week | ● 24-line rotation · typically 4h 10m a week |
| FF19 | Successive start times varying by more than 2 hours (definition to confirm) Circadian | $\Delta$ 6                                  | $\Delta$ 8                                   |
| FF6  | Night shift over 10h Night shifts                                                     | --                                          | --                                           |
| FF8  | Less than 2 days rest after consecutive nights Night shifts                           | --                                          | --                                           |
| FF9  | Less than 14h rest in any 24h for night shifts Night shifts                           | --                                          | --                                           |
| FF12 | Only one day rest after night shifts Night shifts                                     | --                                          | --                                           |
| FF14 | More than 4 consecutive nights in a rotating pattern Night shifts                     | --                                          | --                                           |
| FF16 | More than 3 consecutive night shifts over 8h Night shifts                             | --                                          | --                                           |
| FF20 | First night shift Night shifts                                                        | --                                          | --                                           |
| FF13 | Less than 12h rest in any 24h Intervals between duties                                | ✓ 0                                         | ✓ 0                                          |

Present: 4 in the worst case, 3 as rostered (today 5) · standing: 2 · FF2 fires on every 06:20 duty, so it is a property of the station's opening time rather than of any design. **Two readings of a cover week, on the one row they move.** A cover week is worked four days of seven and the link does not say which four, so a run figure is a range. The headline number is the ceiling — the four split day-on-day-off, which supplies no 48-hour break and joins the blocks either side. It is reachable: of the 35 ways to place four duties in seven days, the 10 that leave no two rest days together produce exactly it. Worked as a BLOCK, which is what a cover week looks like on the roster, the three rest days fall together and the week always supplies a break — that is the *as rostered* figure beneath it. Checked on every row: FF11 is the only one where the two differ. Which reading applies is a question for the roster office, not for this sheet. This sheet is an aid to a conversation, not a fatigue risk assessment.

## Where it came from

Checkable rather than reproducible: there is no search behind it, and every figure is computed from the cells

### Where this design came from

**1 • It was re-ordered by hand, not searched.** This rotation arrived as a Word table describing a series of named, deliberate moves against *Weeks 17-18 Swapped*— a same-day duty swap at a time, each one argued in the document. It was read straight into the app's own shape and judged; no grid was annealed, so there is no search to reproduce. What can be reproduced is every figure on these pages.

**2 • The document's central claim was tested first, and it holds.** It states that every duty moved stays on the SAME weekday, so the number and type of shifts available on each day are preserved. Checked against *Weeks 17-18 Swapped*: each day's multiset of duties is **identical**, and so is the hour-by-hour coverage curve — every quarter-hour of the whole week. Nothing was invented, nothing re-timed, and no day lost or gained a person.

**3 • The reading was checksummed before anything was built on it.**

A grid read from a document can carry a silent transcription error, so the duty minutes were totalled against the contract first: Monday to Saturday comes to **42,000** minutes against 20 working lines × 35h = 42,000, exactly. Every one of the document's own 24 stated weekly totals also matches its cells to the minute.

**4 • The judge is the workspace's own** — runDesignChecks, weeklyHours, assessHardLimits, assessFatigue, scoreOrder and calcHourlyCoverage, with nothing typed by hand and nothing softened. Where the design misses one of the December 2026 timetable rules the table opposite says so rather than omitting the row.

### Alternatives measured

| DESIGN                                                                | RUN | FF PRESENT | WKENDS | ONE- TURN | STEP   | WK FIT | SCORE |
|-----------------------------------------------------------------------|-----|------------|--------|-----------|--------|--------|-------|
| TF-24-EXT · 8eef9a13 — Targeted Fatigue Redo (this proposal)          | 9   | 4          | 5      | 4/20      | 4h 10m | 34.4   | —     |
| Same Turns · d15e1b74 — the searched proposal                         | 6   | 1          | 6      | 18/20     | 2h 33m | 46.3   | —     |
| By the Book · 0f14abce — the searched proposal                        | 6   | 0          | 6      | 12/20     | 2h 52m | 30.8   | —     |
| WS-24-EXT — Weeks 17-18 Swapped, the design this re-orders · 0bebb675 | 9   | 5          | 6      | 6/20      | 4h 17m | 34.4   | —     |
| WL-24-EXT — Weekday Lates, as supplied · a52ec588                     | 9   | 5          | 6      | 7/20      | 4h 43m | 38.3   | —     |
| WL4-24-F7 — Weekday Lates 4, searched fatigue-first · f0d403d6        | 7   | 1          | 6      | 10/20     | 2h 34m | 34.2   | —     |
| FT-24-R21 — Fifteen Turns Repaired, searched · b76bf9e1               | 6   | 1          | 6      | 16/20     | 3h 27m | 30.7   | —     |
| Workspace default · 29ec9810 (Dec 2026 table, generated)              | 7   | 5          | 5      | 0/20      | 4h 16m | 30.8   | —     |
| Today's 20-line link (for scale)                                      | 9   | 5          | 4      | 7/16      | 4h 0m  | 44.7   | —     |

**How to read this table:** nothing here was picked by a search — this design was supplied or derived, and the rows are the comparisons it is being judged against. *Wk fit* is how evenly the WEEKDAY cover follows the December 2026 timetable traffic curve (lower is better; today's link scores what it scores), so it says nothing about Saturday or Sunday. **Score** is the search's own feel objective and is blank here, because no search produced this design — a blank is not a bad score. The workspace default is the app's own December 2026 duty table, generated and reordered with every switch on.

## LOAD IT

# The rotation, ready to paste

Line number, then Sunday to Saturday. SP is a cover week. Paste the whole block into Links → Import.

Links → Import, then paste the whole block below. Line number, then Sunday to Saturday; SP is a cover week. The workspace re-runs every check on these pages from the pasted cells, so nothing here has to be taken on trust — and the two designers can edit it there like any other design.

The same rotation is also supplied beside this PDF as Targeted-Fatigue-Redo-TF-24-EXT-import.txt (tab-separated, pastes directly) and Targeted-Fatigue-Redo-TF-24-EXT.json (the app's own format).

|    |             |             |             |             |             |             |             |
|----|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| 1  | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 2  | 14:30-23:25 | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 | RD          | RD          |
| 3  | 07:15-15:45 | RD          | 08:30-16:30 | 06:20-14:20 | 06:20-14:20 | 06:20-14:20 | 06:20-14:20 |
| 4  | RD          | RD          | 06:20-14:20 | 08:30-16:30 | RD          | 13:30-22:00 | 14:45-23:55 |
| 5  | 14:30-23:25 | RD          | 14:00-22:30 | 14:00-22:30 | 14:00-22:30 | 16:25-23:55 | RD          |
| 6  | RD          | 06:20-14:20 | 07:00-15:30 | 07:00-15:30 | 07:00-15:30 | 08:30-17:00 | RD          |
| 7  | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 8  | 07:15-15:45 | 06:20-13:45 | RD          | RD          | 13:30-22:00 | 14:00-22:30 | 13:30-21:00 |
| 9  | RD          | 12:00-20:30 | 12:00-20:30 | 12:00-20:30 | 12:00-20:30 | 12:00-20:30 | RD          |
| 10 | RD          | 14:00-22:30 | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 | 14:45-23:55 |
| 11 | RD          | 07:00-15:30 | 07:00-15:30 | RD          | RD          | 06:20-13:35 | 06:20-14:50 |
| 12 | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 13 | RD          | RD          | 14:00-22:30 | 13:30-22:00 | 14:00-22:30 | 14:00-22:30 | 14:30-22:00 |
| 14 | RD          | 16:25-23:55 | 16:25-23:55 | RD          | RD          | 07:00-15:30 | 06:20-14:50 |
| 15 | 07:15-15:45 | 06:20-13:45 | 06:20-13:35 | 06:20-13:35 | 06:20-13:35 | 07:00-15:30 | RD          |
| 16 | RD          | 08:30-17:00 | 08:30-17:00 | 08:30-17:00 | RD          | RD          | RD          |
| 17 | 10:30-19:00 | 16:25-23:55 | RD          | RD          | 06:20-14:20 | 06:20-14:20 | 06:20-14:50 |
| 18 | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 19 | 14:30-23:25 | RD          | 06:20-13:45 | 06:20-13:45 | 06:20-13:45 | 06:20-13:45 | 08:00-16:00 |
| 20 | 08:30-16:30 | 06:20-13:45 | RD          | RD          | 13:30-22:00 | 16:25-23:55 | 13:30-22:00 |
| 21 | 13:30-22:00 | 13:30-22:00 | RD          | 13:30-22:00 | 13:30-22:00 | 13:00-21:00 | RD          |
| 22 | RD          | 14:00-22:30 | 13:30-22:00 | 13:30-22:00 | 13:30-22:00 | RD          | RD          |
| 23 | RD          | RD          | RD          | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 |
| 24 | RD          | RD          | 06:20-13:45 | 06:20-13:45 | 07:00-15:30 | RD          | 06:20-13:45 |

Prepared 22 September 2026 · re-rendered 24 September 2026. Every figure in this document was computed by the Marylebone Roster app's Links modules (v24.22) from exactly these cells; import them and the workspace's Design checks, hard limits, fatigue factors and coverage cards will restate them.